EXECUTIVE BRIEF

SunGrid Member Copilot

An AI assistant that answers SunGrid members' questions from the cooperative's own policies, and checks rooftop-rebate eligibility with the official service.

THE OBJECTIVE

Accurate answers, any time

Members get answers from SunGrid's own policies, with the source shown

Rebate checks to trust

The official service decides; the Copilot never promises a rebate

More time for support

Routine questions are handled; hard ones go to a person

HOW IT WORKS

From a member's question to a safe answer

A member asks

"Do I qualify for the rooftop rebate?"

The Copilot reads

SunGrid's own policies, and works out what kind of question it is

- **General question:** a clear answer, with the policy it comes from
- **Rebate question:** the official service decides; the Copilot reports it and never guesses
- **Unclear or sensitive:** says so honestly and passes it to a support specialist

Built-in guardrails: answers only from SunGrid's documents · never promises a rebate · says "I don't know" rather than guess · politely declines off-topic questions

RESULTS AND COST

Tested before launch, at a low running cost

HOW IT PERFORMED ON 20 TEST QUESTIONS

18/20 questions handled as expected

0 rebate amounts shown to members who did not qualify

3/3 rebate eligibility checks handled correctly

WHAT IT COSTS TO RUN

$6.40

per 1,000 member questions (AI usage)

10,000 a month	**~$64**
100,000 a month	**~$640**

Pilot investment: six weeks of a small team (lead, associate, half a backend engineer) plus cloud hosting

⚠ **Two misses:** one confident wrong answer, and one question it refused when it could have answered. **The fix:** add a second AI check that compares every answer with the SunGrid documents it used, before it reaches the member.

RISKS AND LIMITATIONS

Key risks and how to manage them

Eligibility service not ready

High

The risk: rebate decisions come from another team's system. If it is late or changes, rebate checks stop working.

Mitigation: agree the interface with that team in week 1 and test it daily. If late, launch without rebate checks.

A wrong answer

Medium

The risk: the AI can misread a policy and answer confidently but wrongly. It happened once in testing.

Mitigation: a second AI check compares every answer with the SunGrid documents it used. Unsure answers go to a person.

Wrong member details

Medium

The risk: today members type in their own tier and income. A mistake gives a wrong eligibility result.

Mitigation: members sign in, and their tier and address come from their SunGrid account.

Policies go out of date

Medium

The risk: the Copilot is only as accurate as SunGrid's documents. An outdated policy gives an outdated answer.

Mitigation: every document has a named owner and review date, and each update is tested before it goes live.

RECOMMENDATION

Go, to a limited pilot: six weeks, then decide

Week 1
Agree and approve

Weeks 2–4
Build safely on AWS

Week 5
Pilot with support staff

Week 6
Go / no-go on the results

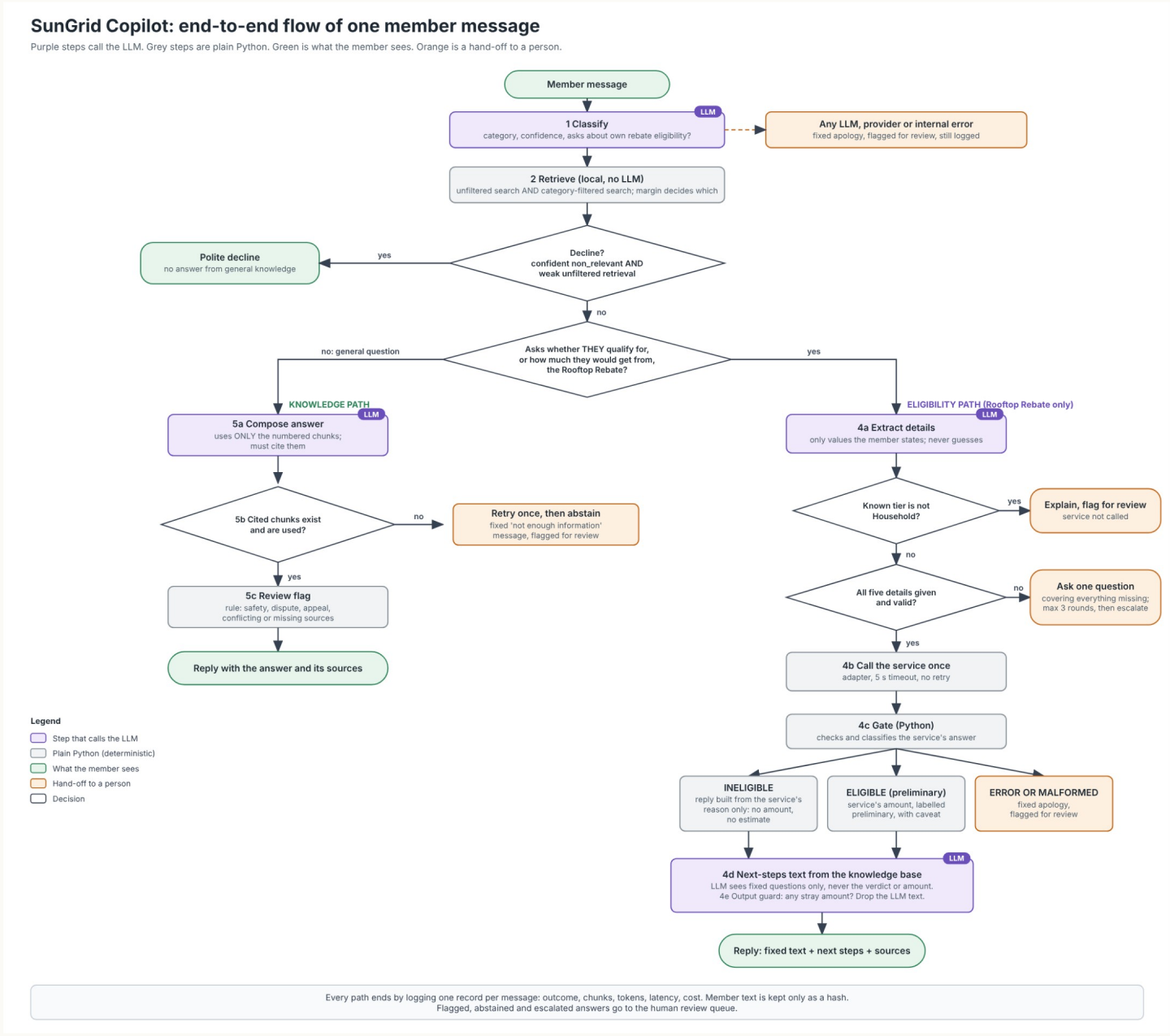
What the pilot delivers

- Sign-in, with tier and address from the member record
- A check that every answer is backed by a policy
- Hard questions passed to a support specialist
- Runs on AWS, monitored, with a cost alarm

How success is measured

- Questions answered without a person
- Answers graded correct by support staff
- Reply time and cost per question
- Rebate amounts given wrongly: must be zero

Needed to start: six-week budget · test access from the eligibility team · AWS and security approval · support staff for the pilot



APPENDIX A

End-to-end flow of one member message

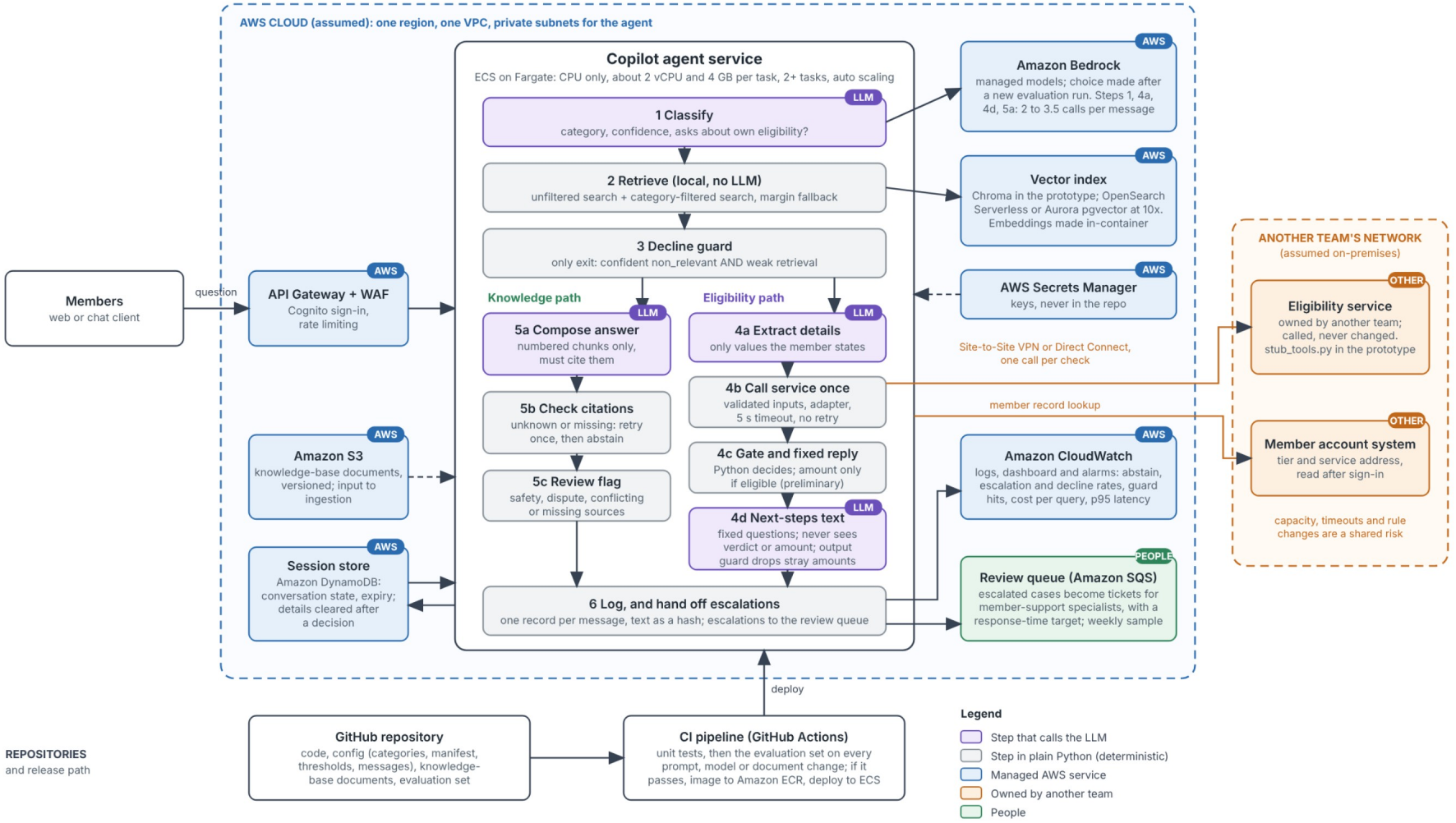
- **Classify** the question (AI)
- **Search** filtered and unfiltered; code picks
- **Decline** only if clearly off-topic
- **Knowledge path:** answer from cited chunks, or abstain
- **Eligibility path:** official service decides, Python gate enforces
- **Every message logged;** hard cases go to a person

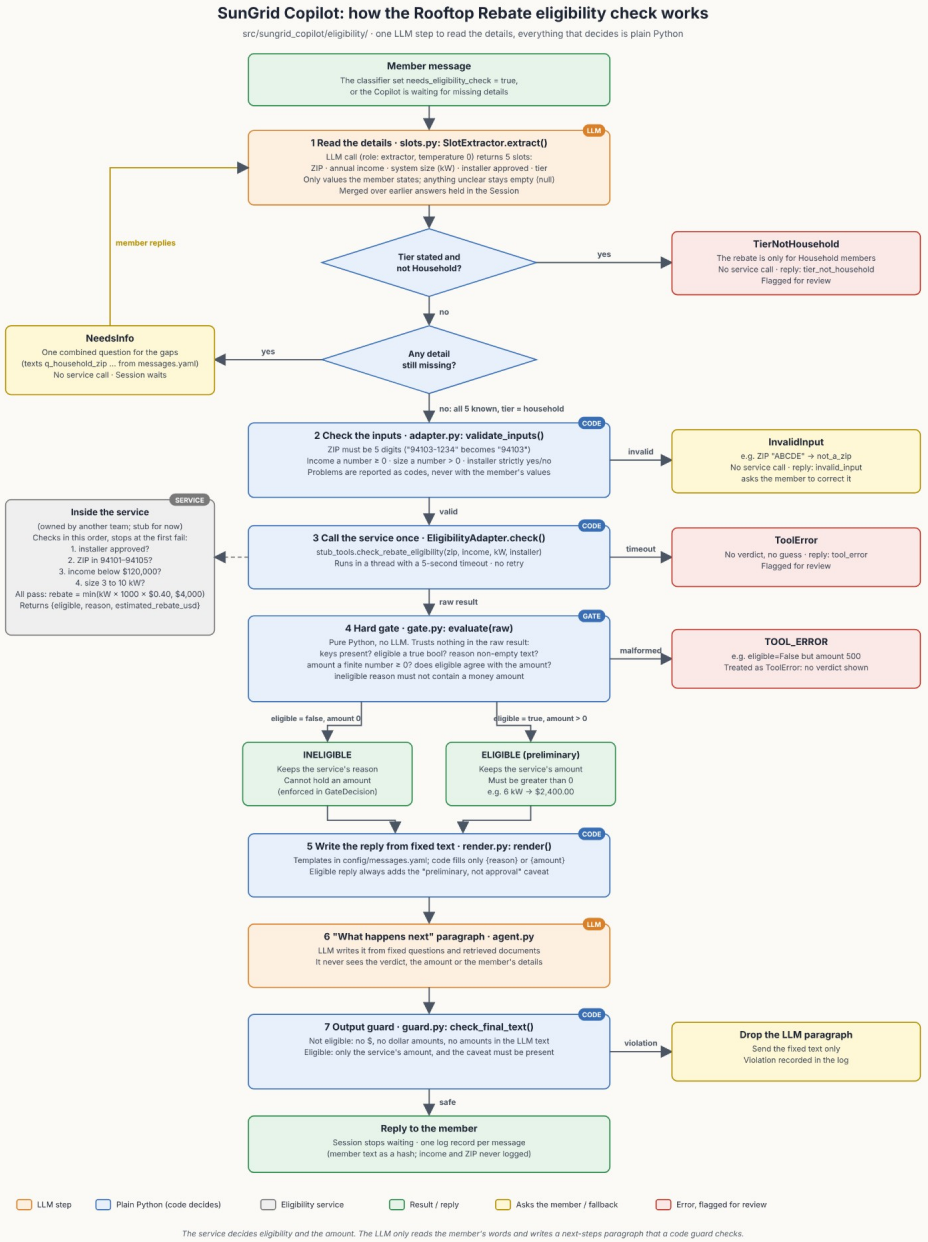
APPENDIX B

Proposed production architecture on AWS

SunGrid Copilot: pipeline and infrastructure

Proposed production layout on AWS. The prototype runs the same flow on one machine, against Azure OpenAI, with a stub in place of the eligibility service.





The service decides eligibility and the amount. The LLM only reads the member's words and writes a next-steps paragraph that a code guard checks.

APPENDIX C

How the rebate rule is enforced in code

- **The AI only reads** the member's words into five details; anything unclear is left empty and asked
- **Inputs are checked,** then the official service is called once (5-second timeout, no retry)
- **A Python gate** turns its answer into eligible, ineligible or error; only "eligible" can carry an amount
- **Fixed reply text** from configuration, then a final guard: no amount can reach an ineligible member
- **Anything odd** (timeout, crash, contradictory answer) means "can't check right now", never a guess

APPENDIX D

Evaluation: 20 questions, automatic checks only

Check	Result
Right topic chosen	20/20
Expected documents found	15/15
Expected source cited	14/14
Right outcome	19/20
Service called only when it should be	20/20
Flagged for review when expected	1/2
No amount where forbidden	20/20
Passed every check	18/20

Miss 1: a confident wrong answer

Question: “Can a Community-tier member run for the board?”

It said: “No, only Household members can.” The documents only say who *votes*, not who can *stand*.

Fix: a second check that every answer is backed by its sources.

Miss 2: refused when it could answer

Question: “How many members and how much capacity does SunGrid have now?”

It said: “I don’t have enough information.” The annual report has the figures: just over 5,000 members, 38 MW.

Fix: clearer answering rules: give the figure and say which period it covers.

Next steps: answers graded by people, repeat runs, and a larger set of questions.

APPENDIX E

What I would improve next

1. Check answers against their sources · High

A second check that every claim is backed by the documents it cites

2. Conversation memory · High

Store each conversation; follow-ups keep their context; personal details cleared

3. Block unofficial rebate amounts · High

Code removes any amount in a general answer that is not in the documents

4. Hand-off to support staff · High

Review queue with a ticket, member contact and the conversation

5. Stronger evaluation · High

Answers graded by people, repeat runs, 100+ questions written blind

6. Update only what changed · Medium

Chunk id = document + content hash: no full rebuild, no duplicates

7. Keyword + meaning search · Medium

Finds exact codes such as “E07”; reranker as documents grow

8. Faster, cheaper replies · Medium

Smaller model for simple steps, structured output, cache common answers

9. LangGraph, when the flow grows · Medium

Durable state and human-approval steps once there are several tools

10. Production basics · Medium

Sign-in and member record, chat widget, CI pipeline, audit log of decisions